SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

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ASSESSMENT TOOL 2 – ANSWERS

Course Name: Artificial Intelligence Course Code: CSA17

Constraint-Based Problem Solving

1. Backtracking Search for Doctor Shift Assignment

Let the variables be the three shifts: Morning, Afternoon, and Night. The values are doctors D1, D2, and D3, and each doctor must be assigned to exactly one shift. First, assign D2 to Morning. This does not violate any constraint because D2 is allowed to work in the Morning shift. Next, assign D1 to Afternoon. This is valid because D1 is not assigned to the Night shift. Finally, assign D3 to Night. This is valid because D3 is not assigned to the Morning shift, and D2 works before D3. The check for the “one doctor per shift” constraint is also satisfied because every shift contains only one doctor. Therefore, all three doctors have exactly one shift and all constraints are satisfied. The final valid schedule is: Morning – D2; Afternoon – D1; Night – D3.

Backtracking constraint checks: (1) D2 = Morning: valid. (2) D1 = Afternoon: valid because D1 is not on Night and the shift is unoccupied. (3) D3 = Night: valid because D3 is not on Morning, D2 is earlier than D3, and the shift is unoccupied. Since a complete valid assignment is found, the search stops without needing to backtrack.

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ASSESSMENT TOOL 2 – ANSWERS

2. BFS Search in a 5 × 5 Grid Using Manhattan Distance

Breadth-First Search (BFS) explores the grid level by level, starting from S. The robot checks the neighbouring cells in the allowed order Up, Down, Left, and Right, adds each unvisited and non-obstacle cell to the queue, and records its parent so that the path can be reconstructed when G is reached. Each move costs 1, so BFS finds a path with the minimum number of moves when all moves have equal cost. The Manhattan distance from a cell (x, y) to the goal (xg, yg) is $|x - xg| + |y - yg|$. This heuristic estimates the number of horizontal and vertical moves to the goal, but standard BFS does not use a priority queue or the heuristic to choose nodes; a heuristic-priority method would be Greedy Best-First Search or A*.

Step-by-step procedure: place S in the queue and mark it visited; remove the front cell and expand its valid neighbours; add each newly discovered neighbour to the rear of the queue and store its parent; repeat until G is removed from the queue or discovered. Then follow the parent links backward from G to S to obtain the optimal path. The path cost is the number of moves in that path, because every move has cost 1.

The provided assessment page does not show the locations of S, G, or the obstacles in the 5 × 5 grid. Therefore, the exact node-expansion order, queue contents, optimal path, and numerical cost cannot be calculated from the supplied information alone. The grid with the start, goal, and obstacle cells is needed to complete those specific steps without making assumptions.

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ASSESSMENT TOOL 2 – ANSWERS

3. Autonomous Rescue Robot in a Partially Observable Environment

a) Constraint analysis

The robot can move only Up, Down, Left, or Right, so diagonal movement is not permitted. Obstacles make some cells unavailable and must never be entered. The robot can observe only adjacent cells, which means it does not initially know the complete layout and must make decisions using limited information. Every ordinary movement costs 1, while entering a risky zone adds 2 more, making the total cost of that move 3. Therefore, the robot must consider both the distance to the survivor and the safety-related extra cost. Since new information is obtained while moving, the robot must update its internal map and revise its route when it discovers a blocked or risky cell.

b) Search strategy and steps

Uniform Cost Search (UCS) is suitable when the robot knows the costs of the transitions it has discovered and needs to find a minimum-cost route. The robot begins at S with cost 0 and stores frontier states in a priority queue ordered by accumulated path cost. It senses adjacent cells, rejects blocked cells, and calculates the cost of entering each available cell: 1 for a normal cell and 3 for a risky cell. It updates the best known cost and parent of a cell whenever a cheaper route is found. The robot expands the frontier cell with the lowest accumulated cost and continues sensing and updating its map. When G is reached and selected for expansion at the lowest cost, the parent links provide the minimum-cost route using the known map. Because the environment is only partially observable, an online-search agent may need to execute a safe move, observe the next cells, update its knowledge, and plan again. If new information changes the map, it should re-evaluate the route.

c) Application of AI concepts

The robot is an intelligent agent because it senses its surroundings, processes information, chooses actions, and acts to reach the survivor. Rationality means selecting an action that best achieves the goal while minimizing travel cost and avoiding known hazards, based on the information currently available. The environment is partially observable because the robot can see only adjacent cells; it is sequential because each move affects later choices; it is dynamic from the robot's perspective because its knowledge changes as it explores; and it is discrete because movement occurs between grid cells using a limited set of actions. The environment also contains costs and obstacles, so the robot must balance progress with safe navigation.

d) Justification

UCS is appropriate because movement costs are not all equal: entering a risky zone costs more than entering an ordinary cell. BFS minimizes the number of moves but does not necessarily minimize total cost when step costs differ. UCS expands the lowest-cost known route first and can find a minimum-total-cost path when the relevant map and transition costs are known. Since the robot initially has only local observations, combining UCS with online sensing and replanning allows it to respond to newly discovered obstacles and risks. This approach supports cost-efficient navigation while avoiding blocked cells and accounting for risky areas.